

Project Documentation: Property Management System (Housing System)

1. Overview

The Property Management System, also referred to as the Housing System, is a comprehensive web-based application designed to streamline the management of rental properties. It provides a decoupled architecture with a robust backend API and a modern frontend interface, allowing property managers (Admins) and renters (Tenants) to interact through dedicated portals.

2. System Architecture

The application follows a client-server architecture:

- **Frontend:** A high-performance web application built using React, Vite, and Tailwind CSS. It focuses on an optimistic UI experience, utilizing contexts for state management (AuthContext, StoreContext).
- **Backend:** A RESTful API developed with Node.js, Express, and MongoDB (via Mongoose). It handles data persistence, authentication, and business logic.

3. Key Features

3.1 Role-Based Access Control (RBAC) & Account Rules

The system supports two primary roles with specific identity constraints:

- **Admin (Landlord):** Full access to the system, including financial auditing, vendor management, and global property configuration.
- **Tenant:** Restricted access to a personal portal where they can view their lease, pay invoices, and submit maintenance requests.

Identity & Authentication Constraints:

- **Role Isolation:** A user cannot share the same email address for both a Tenant account and a Landlord account. Each role requires a unique email.
- **Multi-Property Tenancy:** A single Tenant email can be associated with multiple rental agreements from different landlords. A single tenant account can rent a maximum of 5 different houses.

3.2 Core Modules

Module	Description
Property & Unit Management	Manage properties and their constituent units with cascading deletion support.
Tenant & Lease Tracking	Track tenant details and active lease agreements, including start/end dates.
Financial Management	Handle invoices, process payments, apply late fees, and track overall property expenses.
Maintenance System	Tenants can create maintenance requests; Admins can update status and track resolution.
Communication	Integrated messaging system for tenant-admin interaction and a global announcement board.
Administrative Tools	Management of property owners, external vendors, legal documents, and a comprehensive audit log.

4. Technical Specifications

4.1 Backend Technology Stack

- Runtime: Node.js (v18+)
- Framework: Express.js
- Database: MongoDB (v6+)
- ORM: Mongoose
- Authentication: JWT (JSON Web Tokens)
- Encryption: bcryptjs

4.2 Frontend Technology Stack

- Library: React (v18)
- Build Tool: Vite
- Styling: Tailwind CSS
- Animations: Framer Motion
- Data Visualization: Recharts
- Utilities: date-fns, Lucide React, jsPDF

4.3 API Endpoints

The API is hosted at `http://localhost:5000/api` (local) or the live Render URL.

- Auth: `/auth/login`, `/auth/register`, `/auth/me`
- Resources: `/properties`, `/units`, `/tenants`, `/leases`, `/invoices`, `/expenses`, `/maintenance`, `/messages`, `/owners`, `/vendors`, `/documents`, `/payments`, `/notifications`, `/announcements`, `/audit`

5. Deployment & Access

Live URL: <https://property-management-six-mu.vercel.app/login>

GitHub Repository: <https://github.com/udommiracle2/Property-Management.git>

6. contributors

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